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Applied Calculus IV

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Higher-Order Differential Equations



- 4.1** Preliminary Theory—Linear Equations
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A Linear Differential Equation

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

$$y(x_0) = y_0, \quad y'(x_0) = y_1, \dots, y^{(n-1)}(x_0) = y_{n-1}.$$

An nth-order Initial-Value Problem (IVP)

First-order Initial-Value Problem

$$a_1(x) \neq 0 \quad \leftarrow a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

$$y(x_0) = y_0$$

Second-order Initial-Value Problem

$$a_2(x) \neq 0 \quad \leftarrow a_2(x) \frac{d^2y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

$$y(x_0) = y_0, \quad y'(x_0) = y_1$$

THEOREM 4.1.1 Existence of a Unique Solution

Let $a_n(x)$, $a_{n-1}(x)$, $\dots$, $a_1(x)$, $a_0(x)$ and $g(x)$ be continuous on an interval I and let $a_n(x) \neq 0$ for every x in this interval. If $x = x_0$ is any point in this interval, then a solution $y(x)$ of the initial-value problem (1) exists on the interval and is unique.

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

$$y(x_0) = y_0, \quad y'(x_0) = y_1, \dots, y^{(n-1)}(x_0) = y_{n-1}.$$

Example: verify that $y = 0$ is the unique solution of

$$3y''' + 5y'' - y' + 7y = 0, \quad y(1) = 0, \quad y'(1) = 0, \quad y''(1) = 0$$

Solution:

$a_3(x) \neq 0$ and continuous

Also there are three conditions at point $x = 1$ which belongs the interval of definition

According to the theorem 4.1.1, this differential equation has only one solution

Example: verify that $y = 3e^{2x} + e^{-2x} - 3x$ is the unique solution of

$$y'' - 4y = 12x, \quad y(0) = 4, \quad y'(0) = 1.$$

Solution:

1. y satisfies the DE

$$\begin{aligned} y'' - 4y &= (12e^{2x} + 4e^{-2x}) - 4(3e^{2x} + e^{-2x} - 3x) \\ &= 12e^{2x} + 4e^{-2x} - 12e^{2x} - 4e^{-2x} + 12x \\ &= 12x \end{aligned}$$

2. Also $a_2(x) \neq 0$ and continuous

3. there are two conditions at point $x = 0$ which belongs the interval of definition, and match the solution

$$y(0) = 3e^0 + e^0 - 3 \times 0 = 4, \text{ matches } y(0) = 4.$$

$$y'(0) = 6e^0 - 2e^0 - 3 = 1, \text{ matches } y'(0) = 1.$$

❖ According to the theorem 4.1.1, this differential equation has only one solution

Example: Investigate if $y = cx^2 + x + 3$ is a (unique) solution of

$$x^2y'' - 2xy' + 2y = 6, \quad y(0) = 3, \quad y'(0) = 1$$

Solution:

1. Satisfies the DE

$$\begin{aligned} x^2(2c) - 2x(2cx + 1) + 2(cx^2 + x + 3) &= 2cx^2 - 4cx^2 - 2x + 2cx^2 + 2x + 6 \\ &= (2c - 4c + 2c)x^2 + (-2x + 2x) + 6 \\ &= 6 \end{aligned}$$

2. there are two conditions at point $x = 0$ which belongs the interval of definition, and match the solution

$$y(0): y(0) = c \times 0^2 + 0 + 3 = 3.$$

$$y'(0): y'(0) = 2c \times 0 + 1 = 1.$$

❖ y is a solution of the DE but there might be more solutions for this DE.

According to the theorem 4.1.1, since $a_2(x) = 0$ at $x = 0$, conditions for a unique solution are not fulfilled.

Exercises 4.1

In Problems 1–4 the given family of functions is the general solution of the differential equation on the indicated interval. Find a member of the family that is a solution of the initial-value problem.

1. $y = c_1 e^x + c_2 e^{-x}, (-\infty, \infty);$

$$y'' - y = 0, \quad y(0) = 0, \quad y'(0) = 1$$

2. $y = c_1 e^{4x} + c_2 e^{-x}, (-\infty, \infty);$

$$y'' - 3y' - 4y = 0, \quad y(0) = 1, \quad y'(0) = 2$$

3. $y = c_1 x + c_2 x \ln x, (0, \infty);$

$$x^2 y'' - x y' + y = 0, \quad y(1) = 3, \quad y'(1) = -1$$

4. $y = c_1 + c_2 \cos x + c_3 \sin x, (-\infty, \infty);$

$$y''' + y' = 0, \quad y(\pi) = 0, \quad y'(\pi) = 2, \quad y''(\pi) = -1$$

Second-order Boundary-Value Problem(BVP)

$$a_2(x) \frac{d^2y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

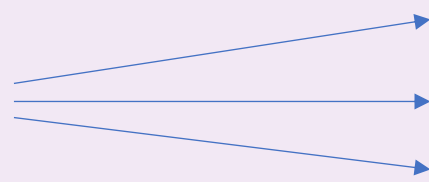
$$y(a) = y_0, \quad y(b) = y_1$$

Conditions are defined at two points

In general,

$$\begin{aligned} \alpha_1 y(a) + \beta_1 y'(a) &= \gamma_1 \\ \alpha_2 y(b) + \beta_2 y'(b) &= \gamma_2. \end{aligned}$$

For example:


$$\begin{aligned} y'(a) &= y_0, & y(b) &= y_1 \\ y(a) &= y_0, & y'(b) &= y_1 \\ y'(a) &= y_0, & y'(b) &= y_1, \end{aligned}$$

Attention

A BVP can have

- Many solutions
- One solution
- No Solution

Provide an example to show it.

Example: $x'' + 16x = 0$ with the family solution $x = c_1 \cos 4t + c_2 \sin 4t$

First, we verify that $x = c_1 \cos 4t + c_2 \sin 4t$ is a general solution of DE

$$\begin{aligned} x'' + 16x &= (-16c_1 \cos 4t - 16c_2 \sin 4t) + 16(c_1 \cos 4t + c_2 \sin 4t) \\ &= -16c_1 \cos 4t - 16c_2 \sin 4t + 16c_1 \cos 4t + 16c_2 \sin 4t \\ &= 0 \end{aligned}$$

We can continue the discussion visually or algebraically

Algebraic Approach: Couple the differential equation with different boundary conditions and find c_1 & c_2

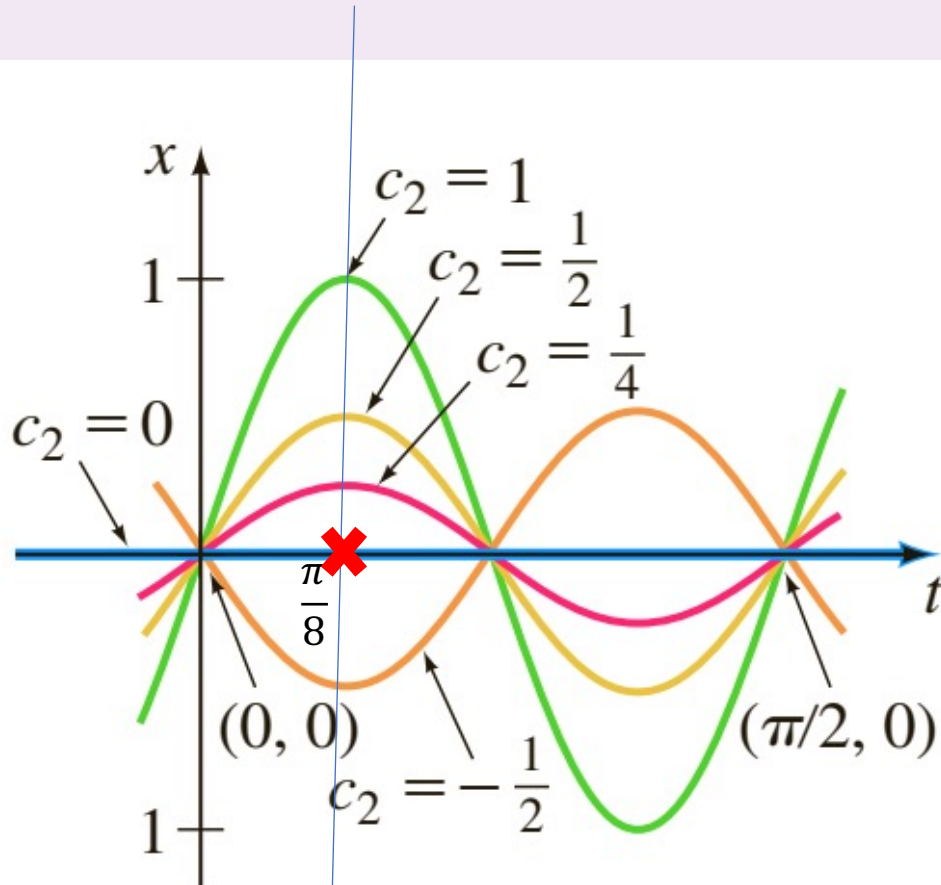
$$x'' + 16x = 0, \quad x(0) = 0, \quad x\left(\frac{\pi}{2}\right) = 0 \quad \Rightarrow \quad \begin{cases} c_1 \cos 0 + c_2 \sin 0 = 0 \\ c_1 \cos 2\pi + c_2 \sin 2\pi = 0 \end{cases} \quad \text{Many solutions}$$

$$x'' + 16x = 0, \quad x(0) = 0, \quad x\left(\frac{\pi}{8}\right) = 0, \quad \Rightarrow \quad \begin{cases} c_1 \cos 0 + c_2 \sin 0 = 0 \\ c_1 \cos \frac{\pi}{2} + c_2 \sin \frac{\pi}{2} = 0 \end{cases} \quad c_1 = 0 \text{ \& } c_2 = 0 \rightarrow x(t) = 0$$

Only one solution

$$x'' + 16x = 0, \quad x(0) = 0, \quad x\left(\frac{\pi}{2}\right) = 1, \quad \Rightarrow \quad \begin{cases} c_1 \cos 0 + c_2 \sin 0 = 0 \\ c_1 \cos 2\pi + c_2 \sin 2\pi = 1 \end{cases} \quad c_1 = 0 \text{ \& } c_1 = 1 \rightarrow \text{No solution}$$

Visual experiment: Alternatively, to build an example, we can draw some solutions (for instance for $c_1 = 0$ and $c_2 = 0, 1, \pm \frac{1}{2}, \frac{1}{4}, \dots$). We can see that



$$x'' + 16x = 0, \quad x(0) = 0, \quad x\left(\frac{\pi}{2}\right) = 0 \quad \text{Multiple solutions}$$

$$x'' + 16x = 0, \quad x(0) = 0, \quad x\left(\frac{\pi}{8}\right) = 0, \quad \text{One solution} \\ x(t) = 0$$

$$x'' + 16x = 0, \quad x(0) = 0, \quad x\left(\frac{\pi}{2}\right) = 1, \quad \text{No solution}$$

Exercises 4.1

In Problems 13 and 14 the given two-parameter family is a solution of the indicated differential equation on the interval $(-\infty, \infty)$. Determine whether a member of the family can be found that satisfies the boundary conditions.

13. $y = c_1 e^x \cos x + c_2 e^x \sin x; \quad y'' - 2y' + 2y = 0$

(a) $y(0) = 1, \quad y'(\pi) = 0$

(b) $y(0) = 1, \quad y(\pi) = -1$

(c) $y(0) = 1, \quad y(\pi/2) = 1$

(d) $y(0) = 0, \quad y(\pi) = 0.$

14. $y = c_1 x^2 + c_2 x^4 + 3; \quad x^2 y'' - 5xy' + 8y = 24$

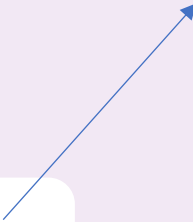
(a) $y(-1) = 0, \quad y(1) = 4$

(b) $y(0) = 1, \quad y(1) = 2$

(c) $y(0) = 3, \quad y(1) = 0$

(d) $y(1) = 3, \quad y(2) = 15$

It always possesses the trivial solution $y = 0$.



$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

Homogeneous

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x),$$

Non-homogeneous

Superposition Principle (homogeneous equations)

Let $y_1, y_2, \dots, y_k$ be k solution of the following homogenous DE on interval I

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

Then, the linear combination of $y_1, y_2, \dots, y_k$ is also a solution (Theorem)

Briefly, $y_1, y_2, \dots, y_k \rightarrow y = c_1 y_1 + c_2 y_2 + \dots + c_k y_k$

Recall

DEFINITION 4.1.1 Linear Dependence/Independence

A set of functions $f_1(x), f_2(x), \dots, f_n(x)$ is said to be **linearly dependent** on an interval I if there exist constants $c_1, c_2, \dots, c_n$, not all zero, such that

$$c_1 f_1(x) + c_2 f_2(x) + \cdots + c_n f_n(x) = 0$$

for every x in the interval. If the set of functions is not linearly dependent on the interval, it is said to be **linearly independent**.

In other words, a set of functions is linearly independent on an interval I if the only constants for which

$$c_1 f_1(x) + c_2 f_2(x) + \cdots + c_n f_n(x) = 0$$

for every x in the interval are $c_1 = c_2 = \cdots = c_n = 0$.

Example: Show that this set of functions is linearly dependent on $(-\frac{\pi}{2}, \frac{\pi}{2})$

$$f_1(x) = \cos^2 x, f_2(x) = \sin^2 x, f_3(x) = \sec^2 x, f_4(x) = \tan^2 x$$

Solution:

$$c_1 \cos^2 x + c_2 \sin^2 x + c_3 \sec^2 x + c_4 \tan^2 x = 0$$

If we choose $c_1 = 1, c_2 = 1, c_3 = -1$ & $c_4 = 1$, the linear combination will be zero,

which shows this set of functions is linearly dependent.

Example: This set of functions is linearly dependent on $(0, \infty)$

$$f_1(x) = \sqrt{x} + 5, f_2(x) = \sqrt{x} + 5x, f_3(x) = x - 1, f_4(x) = x^2$$

Solution:

$$c_1 f_1(x) + c_2 f_2(x) + c_3 f_3(x) + c_4 f_4(x) = 0$$

If we choose $c_1 = 1, c_2 = -1, c_3 = 5$ & $c_4 = 0$, the linear combination will be zero,
which shows this set of functions is linearly dependent.

Challenge

what if the set of functions includes more functions?

Or somehow, it is too difficult to guess

We prefer a method which can indentify being linealy dependent/independent without making aguess

DEFINITION 4.1.2 Wronskian

Suppose each of the functions $f_1(x), f_2(x), \dots, f_n(x)$ possesses at least $n - 1$ derivatives. The determinant

$$W(f_1, f_2, \dots, f_n) = \begin{vmatrix} f_1 & f_2 & \dots & f_n \\ f_1' & f_2' & \dots & f_n' \\ \vdots & \vdots & & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \dots & f_n^{(n-1)} \end{vmatrix},$$

where the primes denote derivatives, is called the **Wronskian** of the functions.

$W(f_1, f_2, \dots, f_n) = 0 \quad \longrightarrow \quad$ The set of functions $f_1, f_2, \dots, f_n$ is linealy dependent

$W(f_1, f_2, \dots, f_n) \neq 0 \quad \longrightarrow \quad$ The set of functions $f_1, f_2, \dots, f_n$ is linealy independent

In Problems 15–22 determine whether the given set of functions is linearly independent on the interval $(-\infty, \infty)$.

15. $f_1(x) = x$, $f_2(x) = x^2$, $f_3(x) = 4x - 3x^2$

16. $f_1(x) = 0$, $f_2(x) = x$, $f_3(x) = e^x$

17. $f_1(x) = 5$, $f_2(x) = \cos^2 x$, $f_3(x) = \sin^2 x$

18. $f_1(x) = \cos 2x$, $f_2(x) = 1$, $f_3(x) = \cos^2 x$

19. $f_1(x) = x$, $f_2(x) = x - 1$, $f_3(x) = x + 3$

20. $f_1(x) = 2 + x$, $f_2(x) = 2 + |x|$

21. $f_1(x) = 1 + x$, $f_2(x) = x$, $f_3(x) = x^2$

22. $f_1(x) = e^x$, $f_2(x) = e^{-x}$, $f_3(x) = \sinh x$

Linearly dependent?

Let $y_1, y_2, \dots, y_n$ be n solutions of the following homogeneous DE on interval I

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

The set of $y_1, y_2, \dots, y_n$ is linearly independent $\iff W(y_1, y_2, \dots, y_n) \neq 0$ (Theorem)

If and only If

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

Any set of $y_1, y_2, \dots, y_n$ of n **linearly independent** solutions is said to be a **fundamental set of solutions**

Theorem

There exists a fundamental set of solutions for this homogeneous linear n th-order differential equation on interval of definition.

namely general solution

$$y = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$$

Attention!

Note that “**general solution**” is a technical term. It refers to a family solutions which can produce **all** possible solutions of a DE.

For example, a general solution of n -th order homogenous DE is a linear combination of n solutions $y_1, y_2, \dots, y_n$, which is a set of linearly independent functions. In other words, it is a linear combination of n fundamental solutions.



To build a general solution, we can't miss any of them in the linear combination

Example: Verify that $x = c_1 \cos 4t + c_2 \sin 4t$ is the general solution of $x'' + 16x = 0$

$$\begin{array}{l} x_1 = \cos 4t \\ x_2 = \sin 4t \end{array} \quad \longrightarrow \quad x_1 \text{ and } x_2 \text{ satisfy the DE}$$

$$\text{Also, } W(\sin 4t, \cos 4t) = 4 \neq 0 \quad \longrightarrow \quad x_1 \text{ and } x_2 \text{ are linearly independent}$$

So, x_1 and x_2 are fundamental solutions and $x = c_1 x_1 + c_2 x_2$ is a general solution

In Problems 23–30 verify that the given functions form a fundamental set of solutions of the differential equation on the indicated interval. Form the general solution.

23. $y'' - y' - 12y = 0$; $e^{-3x}, e^{4x}, (-\infty, \infty)$

24. $y'' - 4y = 0$; $\cosh 2x, \sinh 2x, (-\infty, \infty)$

25. $y'' - 2y' + 5y = 0$; $e^x \cos 2x, e^x \sin 2x, (-\infty, \infty)$

26. $4y'' - 4y' + y = 0$; $e^{x/2}, xe^{x/2}, (-\infty, \infty)$

27. $x^2y'' - 6xy' + 12y = 0$; $x^3, x^4, (0, \infty)$

28. $x^2y'' + xy' + y = 0$; $\cos(\ln x), \sin(\ln x), (0, \infty)$

29. $x^3y''' + 6x^2y'' + 4xy' - 4y = 0$; $x, x^{-2}, x^{-2} \ln x, (0, \infty)$

30. $y^{(4)} + y'' = 0$; $1, x, \cos x, \sin x, (-\infty, \infty)$

Fundamental solutions?

Challenge: Find the solution of nonhomogeneous equations ?

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x),$$

?

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

Find the general solution of the homogenous version $y = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$

namely **complementary function**

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x),$$

Find a particular solution of the non-homogenous version y_p

The general solution of the
nonhomogeneous equation



$$y = \left(c_1 y_1 + c_2 y_2 + \cdots + c_n y_n \right) + y_p$$

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x),$$

$$y = y_c + y_p$$

The general solution of the nonhomogeneous equation = **complementary function + any particular solution**

Example: Verify that $x = c_1 \cos 4t + c_2 \sin 4t + \frac{1}{8}t$ is the general solution of $x'' + 16x = 2t$

Step 1

$$x_1 = \cos 4t$$

$$x_2 = \sin 4t$$



x_1 and x_2 satisfy the DE



$$x_c = c_1 x_1 + c_2 x_2$$

is a general solution of
homogenous equation
(complementary)

Step 2

Also, $x_p'' + 16x_p = 0 + 16\left(\frac{1}{8}t\right) = 2t \rightarrow x_p = \frac{1}{8}t$ is a particular solution of non-homogenous equation



$x = c_1 \cos 4t + c_2 \sin 4t + \frac{1}{8}t$ is the general solution of $x'' + 16x = 2t$

In Problems 31–34 verify that the given two-parameter family of functions is the general solution of the nonhomogeneous differential equation on the indicated interval.

31. $y'' - 7y' + 10y = 24e^x;$

$$y = c_1 e^{2x} + c_2 e^{5x} + 6e^x, (-\infty, \infty)$$

32. $y'' + y = \sec x;$

$$y = c_1 \cos x + c_2 \sin x + x \sin x + (\cos x) \ln(\cos x), (-\pi/2, \pi/2)$$

33. $y'' - 4y' + 4y = 2e^{2x} + 4x - 12;$

$$y = c_1 e^{2x} + c_2 x e^{2x} + x^2 e^{2x} + x - 2, (-\infty, \infty)$$

34. $2x^2 y'' + 5xy' + y = x^2 - x;$

$$y = c_1 x^{-1/2} + c_2 x^{-1} + \frac{1}{15}x^2 - \frac{1}{6}x, (0, \infty)$$

The general solution of the
nonhomogeneous equation

Superposition Principle (nonhomogeneous equations)

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g_1(x)$$

with particular solution y_{p_1}

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g_2(x)$$

with particular solution y_{p_2}

$\vdots$

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g_n(x)$$

with particular solution y_{p_n}



Then, for this equation

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g_1(x) + g_2(x) + \cdots + g_n(x).$$

$$y_{p_1} + y_{p_2} + \cdots + y_{p_n}$$

is a particular solution

Example:

$y_{p_1} = -4x^2$ is a particular solution of $y'' - 3y' + 4y = -16x^2 + 24x - 8$,

$y_{p_2} = e^{2x}$ is a particular solution of $y'' - 3y' + 4y = 2e^{2x}$,

$y_{p_3} = xe^x$ is a particular solution of $y'' - 3y' + 4y = 2xe^x - e^x$.

Find a particular solution for $y'' - 3y' + 4y = -16x^2 + 24x - 8 + 2e^{2x} + 2xe^x - e^x$.

$$y'' - 3y' + 4y = \underbrace{-16x^2 + 24x - 8}_{g_1(x)} + \underbrace{2e^{2x}}_{g_2(x)} + \underbrace{2xe^x - e^x}_{g_3(x)}.$$

$$y_p = y_{p_1} + y_{p_2} + y_{p_3}$$

35. (a) Verify that $y_{p_1} = 3e^{2x}$ and $y_{p_2} = x^2 + 3x$ are, respectively, particular solutions of

$$y'' - 6y' + 5y = -9e^{2x}$$

and $y'' - 6y' + 5y = 5x^2 + 3x - 16.$

(b) Use part (a) to find particular solutions of

$$y'' - 6y' + 5y = 5x^2 + 3x - 16 - 9e^{2x}$$

and $y'' - 6y' + 5y = -10x^2 - 6x + 32 + e^{2x}.$

Solution (a)

For y_{p_1} :

$$\circ y'_{p_1} = 6e^{2x}$$

$$\circ y''_{p_1} = 12e^{2x}$$

$$\circ y''_{p_1} - 6y'_{p_1} + 5y_{p_1} = 12e^{2x} - 36e^{2x} + 15e^{2x} = -9e^{2x} \quad y_{p_1} \text{ satisfies the equation}$$

For y_{p_2} :

$$\circ y'_{p_2} = 2x + 3$$

$$\circ y''_{p_2} = 2$$

$$\circ y''_{p_2} - 6y'_{p_2} + 5y_{p_2} = 2 - 6(2x + 3) + 5(x^2 + 3x) = 5x^2 + 3x - 16 \quad y_{p_2} \text{ satisfies the equation}$$

Solution(b)


$$y'' - 6y' + 5y = \overset{g_1(x)}{\left(5x^2 + 3x - 16\right)} \overset{g_2(x)}{\left(-9e^{2x}\right)}$$
$$y_p = y_{p_1} + y_{p_2} = x^2 + 3x + 3e^{2x}$$

By the superposition principle
(nonhomogeneous equations)

$$y'' - 6y' + 5y = \overset{-2g_1(x)}{\left(-10x^2 - 6x + 32\right)} + \overset{-\frac{1}{9}g_2(x)}{\left(e^{2x}\right)}$$


$$y_p = -2y_{p_2} - \frac{1}{9}y_{p_1} = -2x^2 - 6x - \frac{1}{3}e^{2x}$$

So we can rewrite the Superposition Principle (nonhomogeneous equations) as:

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g_1(x) \quad \text{with particular solution } y_{p_1}$$

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g_2(x) \quad \text{with particular solution } y_{p_2}$$

$\vdots$

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g_n(x) \quad \text{with particular solution } y_{p_n}$$



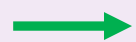
Then, for this equation

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = c_1g_1(x) + c_2g_2(x) + \cdots + c_ng_n(x) \quad c_1y_{p_1} + c_2y_{p_2} + \cdots + c_ny_{p_n}$$

is a particular solution

Later, we will back to these defintions...

Higher-Order Differential Equations



4.1	Preliminary Theory—Linear Equations
4.2	Reduction of Order
4.3	Homogeneous Linear Equations with Constant Coefficients
4.4	Undetermined Coefficients—Superposition Approach
4.5	Undetermined Coefficients—Annihilator Approach
4.6	Variation of Parameters
4.7	Cauchy-Euler Equations
4.8	Green's Functions
4.9	Solving Systems of Linear DEs by Elimination
4.10	Nonlinear Differential Equations

Suppose that we have y_1 , a solution of this second order DE

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$$

Previously, we learnt that the genral solution of a second order DE contains two fundamental solutions

Challenge: How can we find the other fundamental solution?

Reduction of order

Suppose $y_1(x)$ and $y_2(x)$ are linearly independent solutions $\rightarrow \frac{y_2}{y_1}$ is not constant

Let's say $\frac{y_2(x)}{y_1(x)} = u(x)$  $y_2(x) = u(x)y_1(x)$

Now, by replacing $u(x)y_1(x)$ in the DE, we can find $y_2(x)$.

(we literally reduce the order of DE from two to one)

Example: We know that $y_1 = e^x$ is a solution of $y'' - y = 0$. Find the general solution of this DE

Solution

$y_2(x) = u(x)y_1(x) = u(x)e^x$  Replace $y = u(x)e^x$ in the DE

$$y' = ue^x + e^xu', \quad y'' = ue^x + 2e^xu' + e^xu'',$$

$$y'' - y = e^x(u'' + 2u') = 0.$$

$$w = u' \rightarrow w' + 2w = 0 \rightarrow w = c_1 e^{-2x} \rightarrow u' = c_1 e^{-2x}$$

$$\rightarrow u = \frac{1}{2}c_1 e^{-2x} + c_2 \rightarrow y = u(x)e^x = \frac{1}{2}c_1 e^{-x} + c_2 e^x$$

$$\rightarrow (c_1 = 2, c_2 = 0) \quad y_1 = e^{-x}$$

$$y'' + P(x)y' + Q(x)y = 0,$$

y_1 is a solution. Find the general solution of this DE

Replace $y_2(x) = u(x)y_1(x)$

Exercise:

$$y'' + P(x)y' + Q(x)y = 0, \quad \text{Replace } y_2(x) = u(x)y_1(x)$$

$$y' = u'y_1 + uy_1'$$

$$y'' = u''y_1 + 2u'y_1' + uy_1''$$

Replace

$$(u''y_1 + 2u'y_1' + uy_1'') + P(x)(u'y_1 + uy_1') + Q(x)(uy_1) = 0$$

Rearrange

$$u''y_1 + u'(2y_1' + P(x)y_1) + u(y_1'' + P(x)y_1' + Q(x)y_1) = 0$$

$w = u'$

$$u''y_1 + u'(2y_1' + P(x)y_1) = 0 \quad \Rightarrow \quad w'y_1 + w(2y_1' + P(x)y_1) = 0$$

See 2.3

$$w' + \left(2\frac{y_1'}{y_1} + P(x)\right)w = 0$$

$$w' + \left(2\frac{y_1'}{y_1} + P(x)\right) w = 0 \quad \xrightarrow{\text{Integrating factor}} \quad \mu(x) = e^{\int \left(2\frac{y_1'}{y_1} + P(x)\right) dx}$$



$$\int \left(2\frac{y_1'}{y_1} + P(x)\right) dx = 2 \int \frac{y_1'}{y_1} dx + \int P(x) dx$$

$$2 \ln |y_1| + \int P(x) dx = \ln(y_1^2) + \int P(x) dx$$

$$\xrightarrow{\text{Integrating factor}} \quad \mu(x) = e^{\ln(y_1^2) + \int P(x) dx} = y_1^2 \cdot e^{\int P(x) dx}$$

Multiply by IF

$$y_1^2 e^{\int P(x) dx} \cdot w' + y_1^2 e^{\int P(x) dx} \cdot \left(2\frac{y_1'}{y_1} + P(x)\right) w = 0$$

$$\Rightarrow \frac{d}{dx} \left(y_1^2 e^{\int P(x) dx} \cdot w \right) = 0 \quad \Rightarrow y_1^2 e^{\int P(x) dx} \cdot w = c \quad \Rightarrow w = \frac{c}{y_1^2 e^{\int P(x) dx}} = c \cdot \frac{e^{-\int P(x) dx}}{y_1^2}$$

$$w' + \left(2 \frac{y_1'}{y_1} + P(x)\right) w = 0 \quad \longrightarrow \quad w = c_1 \cdot \frac{e^{-\int P(x) dx}}{y_1^2}$$

$$w = u' \quad u(x) = \int w \, dx = c_1 \int \frac{e^{-\int P(x) dx}}{y_1^2} dx + c_2$$

$$y_2(x) = u(x)y_1(x)$$

$$y_2 = y_1(x) \int \frac{e^{-\int P(x) dx}}{y_1^2(x)} dx$$

Example: We know that $y_1 = e^x$ is a solution of $y'' - y = 0$. Find the other solution of this DE

Solution

$$y_2 = y_1(x) \int \frac{e^{-\int P(x)dx}}{y_1^2(x)} dx$$

Use the formula and compare your result with the previous solution, $y_1 = e^{-x}$

Example: $y_1 = x^2$ is a solution of $x^2 y'' - 3xy' + 4y = 0$. Find the general solution of this DE

Solution

$$y_2 = y_1(x) \int \frac{e^{-\int P(x)dx}}{y_1^2(x)} dx$$

Use the formula and then verify your result by replacing in the DE

Exercises 4.2

In Problems 1–16 the indicated function $y_1(x)$ is a solution of the given differential equation. Use reduction of order or formula (5), as instructed, to find a second solution $y_2(x)$.

1. $y'' - 4y' + 4y = 0$; $y_1 = e^{2x}$
2. $y'' + 2y' + y = 0$; $y_1 = xe^{-x}$
3. $y'' + 16y = 0$; $y_1 = \cos 4x$
4. $y'' + 9y = 0$; $y_1 = \sin 3x$
5. $y'' - y = 0$; $y_1 = \cosh x$
6. $y'' - 25y = 0$; $y_1 = e^{5x}$
7. $9y'' - 12y' + 4y = 0$; $y_1 = e^{2x/3}$
8. $6y'' + y' - y = 0$; $y_1 = e^{x/3}$
9. $x^2y'' - 7xy' + 16y = 0$; $y_1 = x^4$
10. $x^2y'' + 2xy' - 6y = 0$; $y_1 = x^2$
11. $xy'' + y' = 0$; $y_1 = \ln x$
12. $4x^2y'' + y = 0$; $y_1 = x^{1/2} \ln x$
13. $x^2y'' - xy' + 2y = 0$; $y_1 = x \sin(\ln x)$
14. $x^2y'' - 3xy' + 5y = 0$; $y_1 = x^2 \cos(\ln x)$

$$1. y'' - 4y' + 4y = 0; \quad y_1 = e^{2x}$$

Define $y = u(x)e^{2x}.$

$$y' = 2ue^{2x} + u'e^{2x},$$

$$y'' = e^{2x}u'' + 4e^{2x}u' + 4e^{2x}u,$$

Replace $y'' - 4y' + 4y = e^{2x}u'' = 0 \Rightarrow u'' = 0 \Rightarrow u = c_1x + c_2.$

$$y = (c_1x + c_2)e^{2x}$$

➡ $c_1 = 1, c_2 = 0, y_2 = xe^{2x}.$

Or use the formula

$$y_2 = y_1(x) \int \frac{e^{-\int P(x)dx}}{y_1^2(x)} dx$$

$$\int P(x)dx = \int (-4)dx = -4x.$$

$$e^{-\int P(x)dx} = e^{-(-4x)} = e^{4x}.$$

Also, $y_1^2(x) = (e^{2x})^2 = e^{4x}.$

$$\int \frac{e^{4x}}{e^{4x}} dx = \int 1dx = x + C.$$

Replace $y_2 = e^{2x} \cdot (x + C).$

$C=0 \quad y_2 = xe^{2x}$

In Problems 17–20 the indicated function $y_1(x)$ is a solution of the associated homogeneous equation. Use the method of reduction of order to find a second solution $y_2(x)$ of the homogeneous equation and a particular solution $y_p(x)$ of the given nonhomogeneous equation.

17. $y'' - 4y = 2; \quad y_1 = e^{-2x}$

18. $y'' + y' = 1; \quad y_1 = 1$

19. $y'' - 3y' + 2y = 5e^{3x}; \quad y_1 = e^x$

20. $y'' - 4y' + 3y = x; \quad y_1 = e^x$

Find y_2 & y_p

17. $y'' - 4y = 2; \quad y_1 = e^{-2x}$

Define $y = u(x)e^{-2x}$

$$y' = -2ue^{-2x} + u'e^{-2x}$$

$$y'' = u''e^{-2x} - 4u'e^{-2x} + 4ue^{-2x}$$

Replace $y'' - 4y = (u'' - 4u')e^{-2x} = 2$

$$w = u', \quad \boxed{w' - 4w = 2e^{2x}}$$

Integrating factor

$\xrightarrow{\hspace{1cm}}$ $e^{-4 \int dx} = e^{-4x}$

Multiply by IF $e^{-4x}w' - 4e^{-4x}w = 2e^{-2x}$

$$\frac{d}{dx}[e^{-4x}w] = 2e^{-2x}$$

$$e^{-4x}w = -e^{-2x} + c_1$$

$$w = u' = -e^{2x} + c_1e^{4x}$$

$$u = -\frac{1}{2}e^{2x} + \frac{1}{4}c_1e^{4x} + c_2$$

$$y = -\frac{1}{2} + \frac{1}{4}c_1e^{2x} + c_2e^{-2x} \quad (c_1 = 4, c_2 = 0)$$

$$y_2 = e^{2x}, \quad y_p = -\frac{1}{2}$$

In Problems 21 and 22 the indicated function $y_1(x)$ is a solution of the given differential equation. Use formula (5) to find a second solution $y_2(x)$ expressed in terms of an integral-defined function. See (iii) in the *Remarks*.

21. $x^2y'' + (x^2 - x)y' + (1 - x)y = 0$; $y_1 = x$

22. $2xy'' - (2x + 1)y' + y = 0$; $y_1 = e^x$

Remark: Sometimes, we can't find the formula for the integral (cannot be expressed in terms of a finite combination of elementary functions). We can express y_2 , in terms of an integral-defined function

$$y_2 = y_1(x) \int_{x_0}^x \frac{e^{-\int_{x_0}^t P(s)ds}}{y_1^2(t)} dt$$

21. $x^2 y'' + (x^2 - x)y' + (1 - x)y = 0; \quad y_1 = x$

Dividing by x^2

$$y'' + \left(1 - \frac{1}{x}\right) y' + \left(\frac{1}{x^2} - \frac{1}{x}\right) y = 0$$

$$P(x) = 1 - \frac{1}{x}$$

Using formula

$$y_2(x) = x \int_{x_0}^x \frac{e^{-(1-1/t)dt}}{t^2} dt = x \int_{x_0}^x \frac{e^{-t+\ln t}}{t^2} dt = x \int_{x_0}^x \frac{e^{-t} e^{\ln t}}{t^2} dt = x \int_{x_0}^x \frac{e^{-t} t}{t^2} dt$$

$$y_2(x) = x \int_{x_0}^x \frac{e^{-t}}{t} dt, \quad x > 0$$

$$y_2 = y_1(x) \int \frac{e^{-\int P(x)dx}}{y_1^2(x)} dx$$

or

$$y_2 = y_1(x) \int_{x_0}^x \frac{e^{-\int_{x_0}^t P(s)ds}}{y_1^2(t)} dt$$